

2079

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Graphics**

Code No: **CSC214** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain the working details of Mid-Point Circle Algorithm? Trace the points along the line path having two end points (6, 9) and (2,3) using Bresenham's line drawing algorithm.
2. Differentiate between object space and image space method of hidden surface removal. Describe the Z-buffer hidden surface removal algorithm.
3. Write the algorithm for Cohen-Sutherland Line clipping. Clip the polygon A (100,150), B (200,250) and C (300,200) with clipping window defined by the coordinates (100,300), (300,300) and (200,100) using Sutherland Hodgeman Polygon Clipping Algorithm.

Section B

Attempt any Eight questions[8×5=40]

4. Reflect a line segment having end points (9,3) and (12,10) about a line $Y = 7$. Draw initial and final result graph as well.
5. Differentiate between Raster and Vector graphics method.
6. Explain about parametric curve. Describe the properties of Bezier curve.
7. What are blobby objects? How it is represented? Explain the wireframe representation of 3D objects.
8. Calculate the total memory required to store a 8 minute video in a SVGA system with 24 bit true color and 25 fps.
9. Find the composite transformation matrix for reflection about a line $y = mx + c$.
10. What is polygon table? Explain the use of this method in 3D object representation.
11. Define the term "Rendering" in computer Graphics. Explain Phong Shading Method with its advantage and disadvantage.
12. Write short notes on (Any TWO) a. BSP Tree b. Virtual Reality c. Intensity Attenuation